In the second slingshot, near Mann's planet, the *Endurance* swings around a suitable IMBH and soars into a gentle rendezvous with the planet: a rendezvous that doesn't require much rocket fuel.

Arrival at Mann's Planet: Ice Clouds

In the movie, the *Endurance* parks in orbit around Mann's planet, and then Cooper and his crew descend to the planet in a Ranger. The planet is covered with ice, as one might expect, since (in my interpretation) it spends most of its life far from the warmth of Gargantua's accretion disk. As the Ranger nears the planet, we see it maneuver among what appear to be clouds, but then it scrapes along one (Figure 19.2) and we discover the cloud is actually made from some sort of ice.



Fig. 19.2. The Ranger scraping the edge of an "ice cloud" on Mann's planet. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

Motivated by a conversation with Paul Franklin, I imagine that these clouds are largely frozen carbon dioxide, "dry ice," and they are starting to be warmed as the planet is on its inward excursion toward the accretion disk, as in Figure 19.1. When warmed, dry ice sublimates—vaporizes—and so what appears to be clouds may be a mixture of dry ice and sublimating vapor; perhaps mostly vapor. At lower altitudes,